

Lecture: Fundamentals of Cyber Security

Cyber Security Goals (CIA Triad), Principles, and Standards

1. Introduction to Cyber Security

Cyber Security is the practice of protecting **Internet-connected systems**, including **hardware, software, networks, and data**, from cyber-attacks, unauthorized access, damage, or disruption.

In today's digital world, cyber threats are rapidly increasing due to the extensive use of computers, mobile devices, cloud platforms, and online services.

Cyber security is not limited to technology alone; it involves:

- **People**
- **Processes**
- **Technology**

These three components work together to reduce threats, manage vulnerabilities, respond to incidents, and ensure system resilience and recovery.

2. Cyber Security Goals – CIA Triad

The **CIA Triad** is the fundamental model of cyber security. It defines the **three primary goals** of information security.

2.1 Confidentiality

Confidentiality ensures that information is **not disclosed to unauthorized individuals, entities, or processes**.

In cyber systems, confidentiality focuses on:

- Protecting sensitive data
- Ensuring privacy of users
- Preventing unauthorized access

Common threats to confidentiality:

- Cracking weak passwords
- Man-in-the-Middle (MITM) attacks

- Phishing attacks
- Poor encryption mechanisms

Measures to ensure confidentiality:

- Data encryption
- Access control mechanisms
- Two-factor authentication (2FA)
- Biometric verification
- Security tokens

Example: Medical records or financial data must be accessible only to authorized personnel.

2.2 Integrity

Integrity refers to maintaining the **accuracy, consistency, and trustworthiness of data** throughout its lifecycle.

It ensures that:

- Data is not altered without authorization
- Any unauthorized modification can be detected

Threats to integrity include:

- Malware attacks
- Unauthorized file modification
- Transmission errors
- Power failures

Measures to ensure integrity:

- Cryptographic checksums
- Hash functions
- File permissions
- Regular data backups
- Uninterrupted Power Supply (UPS)

Example: Bank transaction records must not be modified during transmission or storage.

2.3 Availability

Availability ensures that **authorized users can access information and systems whenever required.**

Even if confidentiality and integrity are preserved, a system is useless if it is not available when needed.

Threats to availability include:

- Denial of Service (DoS) attacks
- Distributed Denial of Service (DDoS) attacks
- Hardware failures
- Natural disasters

Measures to ensure availability:

- Data redundancy
- Regular backups
- Backup power supplies
- Firewalls
- Disaster recovery planning

Example: Online banking services must be available 24×7.

3. Understanding the CIA Triad in Practice

The CIA Triad focuses mainly on **information security**.

While it is the foundation of cyber security, it must be supported by strong **policies, procedures, and standards** to address real-world threats.

Security failures usually occur when:

- One or more CIA components are ignored

- Proper implementation is missing
- Human errors occur

4. Cyber Security Principles

Cyber security principles define **how security controls should be designed and implemented** to achieve CIA goals.

4.1 Defense in Depth

Security should be implemented in **multiple layers**, so that if one layer fails, others still protect the system.

Examples:

- Firewalls
- Intrusion Detection Systems (IDS)
- Authentication mechanisms
- Encryption

4.2 Least Privilege

Users and systems should be granted **only the minimum access required** to perform their tasks.

This principle:

- Reduces attack surface
- Limits damage from compromised accounts

4.3 Accountability

Every action in a system must be:

- Logged
- Audited
- Traceable to a specific user

This helps in:

- Incident investigation

- Cyber forensics
- Legal evidence

4.4 Separation of Duties

Critical operations should be **divided among multiple individuals** to prevent misuse.

Example:

- One person initiates a transaction
- Another person approves it

4.5 Human Layer Security

The uploaded notes emphasize that **humans are the weakest link** in cyber security.

Human-layer controls include:

- Security awareness training
- Phishing simulations
- Access management policies

5. Cyber Security Standards and Frameworks

Cyber security standards provide **guidelines and best practices** to implement security consistently and effectively.

5.1 NIST Cybersecurity Framework (NIST CSF)

Developed by the **National Institute of Standards and Technology (NIST)**.

It provides a structured approach to manage cyber risks.

Core functions of NIST CSF:

1. Identify
2. Protect
3. Detect

4. Respond

5. Recover

NIST focuses on:

- Risk management
- Critical infrastructure protection
- Organizational resilience

5.2 ISO/IEC 27001

ISO/IEC 27001 is an **international standard** for establishing an **Information Security Management System (ISMS)**.

Key features:

- Risk-based approach
- Continuous improvement
- Policy-driven security controls

5.3 ISO/IEC 27002

This standard provides **best-practice security controls** that support ISO/IEC 27001 implementation.

5.4 Legal and Regulatory Standards

The uploaded material highlights the importance of legal compliance:

- **Information Technology Act, 2000 (India)**
- Cybercrime-related sections (43, 66, 66B, 66C, 66D)
- Companies Act, 2013
- National Cyber Security Policy

These standards ensure:

- Legal accountability
- Protection of personal and financial data

- Secure cyber ecosystem

6. Relationship Between Goals, Principles, and Standards

- **CIA Triad** defines *what* needs protection
- **Security Principles** define *how* protection should be applied
- **Security Standards** define *how to implement and enforce* protection

Together, they form the foundation of **effective cyber security**.

7. Conclusion

From the uploaded notes, cyber security fundamentals emphasize that:

- CIA Triad is the core security goal
- Principles guide secure system design
- Standards ensure uniform and legal implementation

A secure cyber environment can only be achieved when **technology, policies, standards, and human awareness** work together.

Lecture: Threats and Threat Actors

Threat Actors / Agents, Attributes of Threat Actors, and Threat Vectors

1. Introduction to Threats in Cyber Security

In cyber security, a **threat** is any **event, action, or circumstance** that has the potential to **harm information systems, networks, or data** by exploiting vulnerabilities.

According to the fundamentals discussed in the uploaded material:

- A **vulnerability** is a weakness in a system
- A **threat** is the potential cause of harm
- An **attack** occurs when a threat is successfully executed

When the **probability of a threat** is combined with the **potential impact**, it is referred to as **risk**.

2. Threat Actors / Threat Agents

A **Threat Actor (or Threat Agent)** is any **individual, group, or entity** that intentionally or unintentionally causes a cyber security incident.

Threat actors can be classified based on **intent, capability, and origin**.

2.1 Insider Threat Actors

Insiders are individuals **within the organization** who have authorized access to systems.

Types of Insider Threats:

- **Malicious insiders** – employees intentionally stealing or damaging data
- **Negligent insiders** – employees who unintentionally cause breaches due to poor security practices
- **Inadvertent insiders** – lack of awareness or training

Examples:

- Employee leaking confidential data
- Weak password usage

- Accidental deletion of critical files

2.2 Outsider Threat Actors

Outsiders operate **outside the organizational boundary**.

Examples:

- Hackers
- Cybercriminals
- Hacktivists
- Nation-state attackers

These attackers usually exploit:

- Network vulnerabilities
- Web application flaws
- Weak authentication mechanisms

2.3 Categories Based on Motivation (from uploaded notes)

a) Political Motivation

- Espionage
- Cyber warfare
- Disrupting national infrastructure
- Making political statements

b) Economic Motivation

- Financial fraud
- Intellectual property theft
- Ransomware attacks
- Credit card fraud

c) Socio-Cultural Motivation

- Ideological goals

- Activism (Hacktivism)
- Curiosity or fame
- Ego gratification

2.4 Other Common Threat Actors

Threat Actor	Description
Script Kiddies	Low-skilled attackers using ready-made tools
Organized Cybercriminals	Highly skilled groups focused on profit
Terrorist Groups	Use cyber attacks to create fear and disruption
Nation-States	Conduct cyber espionage and cyber warfare

3. Attributes of Threat Actors

Attributes of threat actors describe their **capabilities and behavior**, which help security professionals assess risk.

3.1 Motivation

The reason behind an attack:

- Financial gain
- Political influence
- Revenge
- Curiosity
- Sabotage

3.2 Capability (Skill Level)

Threat actors vary in technical expertise:

- Low (script kiddies)
- Medium (organized attackers)
- High (nation-state actors)

Higher capability usually means:

- Sophisticated attacks
- Stealthy techniques
- Use of zero-day vulnerabilities

3.3 Resources

Includes:

- Access to tools and malware
- Infrastructure (botnets, servers)
- Financial backing

Nation-state actors typically possess **high resources**, while individual attackers have limited resources.

3.4 Access Level

- **Insiders** already have system access
- **Outsiders** must first gain unauthorized access

3.5 Intent

Threat actors may be:

- **Intentional** – deliberate attacks
- **Unintentional** – accidental actions (human error)

3.6 Persistence

Some attackers:

- Perform quick attacks
- Maintain long-term presence (Advanced Persistent Threats – APTs)

4. Threat Vectors

A **Threat Vector** is the **path or method used by a threat actor to gain unauthorized access** to a system.

Threat vectors exploit vulnerabilities in:

- Software
- Hardware
- Networks
- Human behavior

4.1 Network-Based Threat Vectors

- Man-in-the-Middle (MITM) attacks
- Denial of Service (DoS / DDoS)
- Packet sniffing
- Session hijacking

4.2 Web-Based Threat Vectors

From the uploaded material:

- SQL Injection
- Code Injection
- File Inclusion attacks
- URL interpretation
- DNS Spoofing

4.3 Malware-Based Threat Vectors

Malware acts as a delivery mechanism for attacks.

Common examples:

- Viruses
- Worms
- Trojans
- Logic bombs
- Backdoors
- Bots

4.4 Email-Based Threat Vectors

- Phishing
- Spear phishing
- Malicious attachments
- Fake links

These attacks primarily exploit the **human layer**, which is highlighted as the weakest link.

4.5 Physical Threat Vectors

- Theft of laptops or mobile devices
- Unauthorized access to hardware
- Hardware tampering
- Counterfeit devices

4.6 Wireless and Mobile Threat Vectors

From your uploaded notes:

- Bluetooth hacking
- Mobile malware

- Overbilling attacks
- Spoofed PDP attacks
- Signaling-level attacks

5. Relationship Between Threat Actors, Attributes, and Threat Vectors

- **Threat actors** decide *who attacks*
- **Attributes** explain *why and how capable they are*
- **Threat vectors** explain *how the attack is carried out*

Effective cyber security requires:

- Identifying threat actors
- Understanding their attributes
- Blocking or mitigating threat vectors



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Lecture 3: Malware and Trojans

PART A: Malware

1. Introduction to Malware

Malware (Malicious Software) is any software intentionally designed to **disrupt, damage, spy on, or gain unauthorized access** to computer systems, networks, or data.

Malware violates one or more goals of the **CIA Triad**:

- Confidentiality (data theft)
- Integrity (data modification)
- Availability (system disruption)

2. Malware Infection

Malware infection refers to the process by which malware enters, spreads, and executes on a target system.

Common Infection Methods

- Email attachments and phishing links
- Malicious websites (drive-by downloads)
- Infected USB drives
- Pirated or cracked software
- Software vulnerabilities

3. Components of Malware

A typical malware consists of the following components:

3.1 Infection Mechanism

Determines **how malware enters the system**

Example: email attachment, exploit kit



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3.2 Payload

The **malicious action** performed by malware

Examples:

- Data theft
- File deletion
- Encryption (ransomware)

3.3 Command and Control (C2)

Enables communication between the malware and the attacker's server.

3.4 Persistence Mechanism

Ensures malware **remains active after reboot**.

3.5 Propagation Mechanism

Allows malware to **spread to other systems**.

4. Types of Malware

4.1 Virus

- Attaches itself to a host file
- Requires user action to execute
- Spreads when infected file is shared

4.2 Worm

- Self-replicating malware
- Spreads without user interaction
- Consumes network bandwidth

4.3 Trojan Horse

- Disguised as legitimate software
- Does not self-replicate
- Creates backdoors



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4.4 Ransomware

- Encrypts victim's files
- Demands ransom payment

4.5 Spyware

- Collects sensitive information
- Monitors user activity

4.6 Adware

- Displays unwanted advertisements
- May track browsing behavior

4.7 Rootkit

- Hides malware presence
- Gains privileged access

4.8 Bot / Botnet

- Turns infected system into a zombie
- Controlled remotely by attacker

PART B: Trojans

5. What is a Trojan?

A **Trojan Horse** is a type of malware that **appears legitimate** but performs **hidden malicious actions** once executed.

Key characteristics:

- Disguised as trusted software
- Requires user execution
- Opens backdoors
- Enables remote control



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6. Indications of Trojan Attack

Common signs of Trojan infection include:

- Slow system performance
- Unknown programs running
- Unexpected pop-up windows
- Disabled antivirus software
- Unusual network traffic
- Unauthorized access attempts

7. How Hackers Use Trojans

Hackers use Trojans to:

- Steal login credentials
- Monitor keystrokes (keyloggers)
- Install additional malware
- Control victim's system remotely
- Participate in botnet attacks

8. Types of Trojans

8.1 Backdoor Trojans: Provide unauthorized remote access

8.2 Remote Access Trojans (RATs): Full control over victim's system

8.3 Banking Trojans: Steal financial credentials

8.4 Downloader Trojans: Download additional malware

8.5 Spy Trojans: Monitor user activities

8.6 Rootkit Trojans: Hide malicious processes

8.7 DDoS Trojans: Launch distributed denial-of-service attacks



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Lecture 4: Viruses, Worms, and Other Malware Types

PART A: Computer Viruses

1. What is a Computer Virus?

A **computer virus** is a type of **malicious software** that **attaches itself to a legitimate host file or program** and spreads when the infected file is executed by the user.

Key Characteristics

- Requires a host file
- Requires user action to activate
- Can replicate and spread
- Can modify or destroy data

2. Purpose of Creating Viruses

Viruses are created for various reasons, including:

1. **Curiosity and Experimentation**
2. **Ego and Fame** (to demonstrate skills)
3. **Financial Gain** (data theft, extortion)
4. **Sabotage** (destroy systems or data)
5. **Political or Ideological Motivation**

3. Indications of Virus Attack

Common symptoms of virus infection include:

- Slow system performance



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- Frequent system crashes
- Unexpected system restarts
- Corrupted or missing files
- Unusual pop-up messages
- Increased disk or CPU usage
- Antivirus alerts or disabled security software

4. Stages of Virus Lifecycle

A computer virus typically goes through **four stages**:

4.1 Dormant Phase

- Virus remains inactive
- Waits for specific conditions (date, event)

4.2 Propagation Phase

- Virus replicates and infects other files or systems

4.3 Triggering Phase

- Virus is activated when a condition is met

4.4 Execution Phase

- Virus payload is executed
- Causes damage or performs malicious activity



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5. Types of Computer Viruses

5.1 File Infector Viruses

- Attach to executable files (.exe, .com)

5.2 Boot Sector Viruses

- Infect boot sector or Master Boot Record (MBR)
- Activate during system startup

5.3 Macro Viruses

- Written in macro languages (e.g., MS Word macros)

5.4 Polymorphic Viruses

- Change code structure to evade detection

5.5 Metamorphic Viruses

- Rewrite their entire code on each infection

5.6 Resident Viruses

- Stay in memory and infect files as they are accessed

5.7 Non-Resident Viruses

- Execute, infect selected files, then terminate



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PART B: Computer Worms

6. How is a Worm Different from a Virus?

Feature	Virus	Worm
Host file required	Yes	No
User interaction	Required	Not required
Self-replication	Yes	Yes
Network spreading	Limited	Extensive

PART C: Other Malware Types

7. Rootkits

A **rootkit** is malware designed to **hide its presence** and provide **privileged (root-level) access** to attackers.

Functions

- Conceal files and processes
- Disable security software
- Maintain persistent access

8. Potentially Unwanted Applications (PUAs)

PUAs are software that:

- Are not explicitly malicious
- Perform unwanted actions
- Often installed unknowingly



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Examples

- Browser toolbars
- Ad injectors
- System optimizers with false alerts

9. Spyware

Spyware secretly **monitors user activities** and collects:

- Browsing history
- Login credentials
- Personal data

10. Keylogger

A **keylogger** records every keystroke typed on a system.

Uses by attackers

- Capture passwords
- Steal credit card details
- Monitor communications

11. Botnets

A **botnet** is a network of **compromised computers (bots or zombies)** controlled by an attacker (botmaster).

Uses

- DDoS attacks
- Spam campaigns
- Credential stuffing
- Cryptomining



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12. Fileless Malware

Fileless malware:

- Does not write files to disk
- Operates in memory
- Uses legitimate tools (PowerShell, Windows Management Instrumentation (WMI))

Advantages for attackers

- Harder to detect
- Bypasses traditional antivirus

13. Summary Table

Malware Type	Key Feature
Virus	Requires host file
Worm	Spreads automatically
Rootkit	Hides presence
PUA	Unwanted but not malicious
Spyware	Data spying
Keylogger	Keystroke capture
Botnet	Remote-controlled network
Fileless Malware	Memory-based attack

MCQs: Fundamentals of Cyber Security

1. Cyber security is mainly concerned with protecting:

- A. Only computer hardware
 - B. Internet-connected systems and digital data
 - C. Electrical power systems only
 - D. Physical buildings
-

2. Cyber security is best described as a combination of:

- A. Software and hardware
 - B. Laws and regulations
 - C. People, processes, and technology
 - D. Networks and protocols
-

3. Which of the following is the primary goal of cyber security?

- A. Faster computation
 - B. Protection of information assets
 - C. Increasing system complexity
 - D. Eliminating the Internet
-

4. The CIA Triad stands for:

- A. Control, Integrity, Access
 - B. Confidentiality, Integrity, Availability
 - C. Cyber, Information, Assurance
 - D. Confidentiality, Inspection, Authorization
-

5. Which CIA component ensures that data is accessible when required?

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authentication
-

6. Encryption mainly supports which security goal?

- A. Availability
 - B. Confidentiality
 - C. Accountability
 - D. Redundancy
-

7. Which of the following is a threat to confidentiality?

- A. Power failure
 - B. Data backup
 - C. Man-in-the-Middle attack
 - D. Data redundancy
-

8. Cryptographic checksums and hash functions are used to ensure:

- A. Availability
 - B. Confidentiality
 - C. Integrity
 - D. Authentication
-

9. Denial-of-Service (DoS) attacks primarily target:

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authorization
-

10. Data redundancy and backup power supplies help ensure:

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Non-repudiation
-

11. Which principle limits user access to the minimum required permissions?

- A. Defense in depth
 - B. Least privilege
 - C. Accountability
 - D. Fail-safe defaults
-

12. The concept of using multiple layers of security controls is known as:

- A. Least privilege
 - B. Separation of duties
 - C. Defense in depth
 - D. Risk management
-

13. Which principle ensures actions can be traced to a specific user?

- A. Confidentiality
 - B. Integrity
 - C. Accountability
 - D. Availability
-

14. Dividing critical responsibilities among multiple individuals follows the principle of:

- A. Accountability
 - B. Defense in depth
 - C. Separation of duties
 - D. Least privilege
-

15. According to cyber security fundamentals, the weakest link in security is:

- A. Hardware
 - B. Software
 - C. Network
 - D. Human factor
-

16. A threat in cyber security is:

- A. A system weakness
- B. A successful attack

- C. A potential cause of harm
 - D. A security control
-

17. A vulnerability is best described as:

- A. An active cyber attack
 - B. A weakness in a system
 - C. A threat actor
 - D. A legal violation
-

18. Which of the following is an example of a web-based attack?

- A. Worm
 - B. Trojan
 - C. SQL Injection
 - D. Botnet
-

19. Malware that replicates itself without a host program is called:

- A. Virus
 - B. Worm
 - C. Trojan
 - D. Logic bomb
-

20. Which standard provides a framework for managing cyber security risks?

- A. PCI-DSS
 - B. ISO/IEC 27001
 - C. NIST Cybersecurity Framework
 - D. GDPR
-

21. The NIST Cybersecurity Framework consists of how many core functions?

- A. Three
- B. Four
- C. Five
- D. Six

22. Which of the following is NOT a NIST CSF core function?

- A. Identify
 - B. Protect
 - C. Detect
 - D. Authenticate
-

23. ISO/IEC 27001 focuses on:

- A. Network performance
 - B. Information Security Management Systems (ISMS)
 - C. Cryptographic algorithms
 - D. Software development
-

24. Cyber security policies are best described as:

- A. Static documents
 - B. Informal guidelines
 - C. Living documents
 - D. One-time rules
-

25. Risk in cyber security is commonly defined as:

- A. Threat + vulnerability
 - B. Vulnerability $\times$ threat
 - C. Threat probability $\times$ potential loss
 - D. Attack $\times$ damage
-

MCQs: Threats and Threat Actors

1. A cyber threat is best defined as:

- A. A weakness in a system
 - B. A successful cyber attack
 - C. A potential cause of harm to an information system
 - D. A security policy
-

2. When a threat successfully exploits a vulnerability, it results in:

- A. Risk
 - B. Attack
 - C. Exposure
 - D. Incident response
-

3. A threat actor is:

- A. Only an external hacker
 - B. A software vulnerability
 - C. Any individual or entity capable of causing a security incident
 - D. A firewall rule
-

4. Which of the following is an insider threat?

- A. Nation-state hacker
 - B. External cybercriminal
 - C. Employee accidentally deleting critical data
 - D. Distributed botnet
-

5. An employee who unintentionally causes a data breach due to lack of awareness is called:

- A. Malicious insider
 - B. Negligent insider
 - C. External attacker
 - D. Hacktivist
-

6. Which threat actor is primarily motivated by financial gain?

- A. Hacktivist
- B. Nation-state

- C. Organized cybercriminal
 - D. Script kiddie
-

7. Cyber attacks aimed at espionage or disrupting national infrastructure are usually motivated by:

- A. Economic factors
 - B. Socio-cultural factors
 - C. Political factors
 - D. Curiosity
-

8. Hackers driven by ideology or publicity are known as:

- A. Cyber terrorists
 - B. Organized criminals
 - C. Hacktivists
 - D. Insiders
-

9. Which of the following is an attribute of a threat actor?

- A. Firewall rules
 - B. Encryption algorithm
 - C. Motivation
 - D. Backup strategy
-

10. The capability of a threat actor refers to:

- A. Legal authority
 - B. Technical skill and expertise
 - C. Physical location
 - D. Organization size
-

11. Nation-state attackers typically have:

- A. Low resources and low skills
- B. High resources and high skills
- C. Low persistence
- D. Limited infrastructure

12. An attacker who maintains long-term access to a system is exhibiting:

- A. Opportunistic behavior
 - B. Low motivation
 - C. Persistence
 - D. Negligence
-

13. A threat vector is:

- A. A security control
 - B. A type of vulnerability
 - C. The path used to carry out an attack
 - D. A cyber law
-

14. Which of the following is a web-based threat vector?

- A. SQL Injection
 - B. Trojan Horse
 - C. Keylogger
 - D. Logic bomb
-

15. Phishing attacks mainly exploit:

- A. Network hardware
 - B. Cryptographic flaws
 - C. Human behavior
 - D. Operating systems
-

16. Which attack intercepts communication between two parties?

- A. DoS attack
 - B. Man-in-the-Middle attack
 - C. Brute-force attack
 - D. Dictionary attack
-

17. Malware that disguises itself as legitimate software is called:

- A. Virus
 - B. Worm
 - C. Trojan Horse
 - D. Bot
-

18. Bluetooth hacking and mobile malware fall under:

- A. Physical threat vectors
 - B. Network threat vectors
 - C. Wireless and mobile threat vectors
 - D. Application threat vectors
-

19. Theft of laptops and hardware tampering are examples of:

- A. Software attacks
 - B. Physical threat vectors
 - C. Network attacks
 - D. Social engineering
-

20. Risk in cyber security is commonly defined as:

- A. Threat + attack
 - B. Vulnerability $\times$ impact
 - C. Threat probability $\times$ potential loss
 - D. Attack $\times$ vulnerability
-

1. Malware is defined as:

- A. Authorized system software
 - B. Software designed to protect data
 - C. Malicious software intended to harm systems
 - D. Hardware failure
-

2. Which CIA triad component is mainly violated by ransomware?

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authentication
-

3. Which malware component performs the actual malicious activity?

- A. Infection mechanism
 - B. Payload
 - C. C2 server
 - D. Persistence module
-

4. Which malware spreads without user interaction?

- A. Virus
 - B. Trojan
 - C. Worm
 - D. Spyware
-

5. A Trojan differs from a virus because it:

- A. Self-replicates
 - B. Attaches to files
 - C. Disguises as legitimate software
 - D. Requires no user interaction
-

6. Which is NOT a sign of Trojan infection?

- A. Slow system performance
 - B. Disabled antivirus
 - C. Increased security updates
 - D. Unusual network traffic
-

7. Trojans mainly enter systems through:

- A. BIOS attacks
- B. Legitimate software downloads

- C. Hardware failures
 - D. Power surges
-

8. Which Trojan allows attackers full remote control?

- A. Downloader Trojan
 - B. RAT
 - C. Spy Trojan
 - D. Banking Trojan
-

9. Which port is commonly used by Back Orifice Trojan?

- A. 80
 - B. 21
 - C. 443
 - D. 31337
-

10. NetBus Trojan commonly uses which port?

- A. 8080
 - B. 12345
 - C. 22
 - D. 3389
-

11. A Trojan that steals banking credentials is called:

- A. Rootkit Trojan
 - B. Spy Trojan
 - C. Banking Trojan
 - D. Downloader Trojan
-

12. Malware that hides its presence is:

- A. Adware
- B. Worm
- C. Rootkit
- D. Virus

13. Botnets are mainly used for:

- A. Data backup
 - B. System updates
 - C. DDoS attacks
 - D. Software optimization
-

14. The main difference between worm and Trojan is:

- A. Worm spreads automatically
 - B. Trojan spreads automatically
 - C. Worm hides itself
 - D. Trojan uses encryption
-

15. Trojans primarily violate which security goal?

- A. Availability
 - B. Confidentiality
 - C. Integrity
 - D. All of the above
-

MCQs: Viruses, Worms & Other Malware

1. A computer virus is a program that:

- A. Spreads automatically without user interaction
 - B. Attaches itself to a host file and spreads when executed
 - C. Operates only in system memory
 - D. Uses only network connections
-

2. Which of the following is a key characteristic of a virus?

- A. Does not require a host
 - B. Requires user action to spread
 - C. Spreads independently
 - D. Exists only in RAM
-

3. The main purpose of creating computer viruses is:

- A. Improving system performance
 - B. Protecting user data
 - C. Causing damage, theft, or disruption
 - D. Ensuring data availability
-

4. Which of the following is NOT an indication of a virus attack?

- A. Slow system performance
 - B. Unexpected system crashes
 - C. Frequent antivirus updates
 - D. Corrupted files
-

5. In which stage of the virus lifecycle does the virus remain inactive?

- A. Propagation phase
 - B. Triggering phase
 - C. Dormant phase
 - D. Execution phase
-

6. The stage in which a virus replicates itself is called:

- A. Dormant phase
 - B. Propagation phase
 - C. Execution phase
 - D. Termination phase
-

7. A virus that infects executable files is called:

- A. Boot sector virus
 - B. Macro virus
 - C. File infector virus
 - D. Polymorphic virus
-

8. Which virus infects the master boot record (MBR)?

- A. File infector virus
 - B. Boot sector virus
 - C. Resident virus
 - D. Macro virus
-

9. A polymorphic virus is designed to:

- A. Stay only in memory
 - B. Change its code to evade detection
 - C. Infect boot sectors only
 - D. Spread without user action
-

10. Which malware does NOT require a host file?

- A. Virus
 - B. Worm
 - C. Macro virus
 - D. File infector virus
-

11. The primary difference between a worm and a virus is that a worm:

- A. Requires user execution
 - B. Needs a host file
 - C. Spreads automatically over networks
 - D. Cannot replicate
-

12. Rootkits are mainly used to:

- A. Display advertisements
 - B. Encrypt files
 - C. Hide malicious activities
 - D. Improve system performance
-

13. Which malware type records keystrokes?

- A. Spyware
- B. Rootkit

- C. Keylogger
 - D. Worm
-

14. Spyware primarily aims to:

- A. Damage system files
 - B. Monitor and collect user information
 - C. Spread over networks
 - D. Encrypt data
-

15. Potentially Unwanted Applications (PUAs) are:

- A. Always malicious
 - B. System viruses
 - C. Unwanted but not strictly malicious software
 - D. Network worms
-

16. A botnet consists of:

- A. Secure computers
 - B. Antivirus servers
 - C. Compromised systems controlled remotely
 - D. Backup systems
-

17. Botnets are commonly used for:

- A. Data backup
 - B. Software updates
 - C. Distributed Denial-of-Service (DDoS) attacks
 - D. System recovery
-

18. Fileless malware primarily operates:

- A. From executable files
- B. From boot sectors
- C. In system memory
- D. From external storage

19. Fileless malware is difficult to detect because it:

- A. Uses encryption
- B. Writes files to disk
- C. Avoids traditional file-based detection
- D. Infects only hardware

20. Which malware uses legitimate system tools like PowerShell to execute attacks?

- A. Virus
- B. Worm
- C. Fileless malware
- D. Adware



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Lecture 5: Vulnerabilities in Cyber Security

1. What is a Vulnerability?

A **vulnerability** is a **weakness, flaw, or gap** in a system, network, application, or process that can be **exploited by a threat actor** to compromise security.

Vulnerabilities may arise from:

- Design flaws
- Implementation errors
- Misconfigurations
- Human mistakes

A vulnerability by itself does not cause harm; **harm occurs when a threat exploits it.**

2. Common Areas of Vulnerability

2.1 Hardware Vulnerabilities

- **Outdated devices** are systems running old hardware or software that no longer receive security updates or vendor support.
- **Physical access to systems** means an unauthorized person can **touch, open, connect to, or control** a device directly.
- **Lack of hardware security controls** This means missing built-in hardware protections that defend systems against tampering, unauthorized access, and low-level attacks.



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2.2 Software Vulnerabilities

- **Bugs in applications**
- **An unpatched operating system** is one that has not received or installed the latest security updates released by the vendor (e.g., Microsoft, Linux distributions, Apple).
- **Insecure code** means programming code that has weaknesses or flaws that attackers can exploit to break into a system, steal data, or cause damage.

2.3 Network Vulnerabilities

- Misconfigured firewalls
- Insecure protocols (FTP, Telnet)
- Poor network monitoring

2.4 Human Vulnerabilities

- Weak passwords
- Social engineering attacks
- Lack of security awareness

3. Impact Caused Due to Vulnerabilities

If vulnerabilities are exploited, the following impacts may occur:

- Data breaches and information leakage
- Financial loss
- Service disruption
- Loss of customer trust and reputation
- Legal and regulatory penalties



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4. Risk in Cyber Security

In cybersecurity, **risk** refers to the probability that a threat will successfully exploit a system vulnerability.

Risk management involves:

- Identifying vulnerabilities
- Assessing threats
- Implementing controls
- Continuous monitoring

5. Types of Vulnerabilities: Vulnerabilities can be classified into:

a) Technical Vulnerabilities

- Software bugs
- Weak encryption
- Misconfigurations

b) Administrative Vulnerabilities

- Poor security policies
- Inadequate training

c) Physical Vulnerabilities

physical environment (buildings, rooms, equipment areas) is not properly protected, making it easier for unauthorized people to access systems or sensitive information.